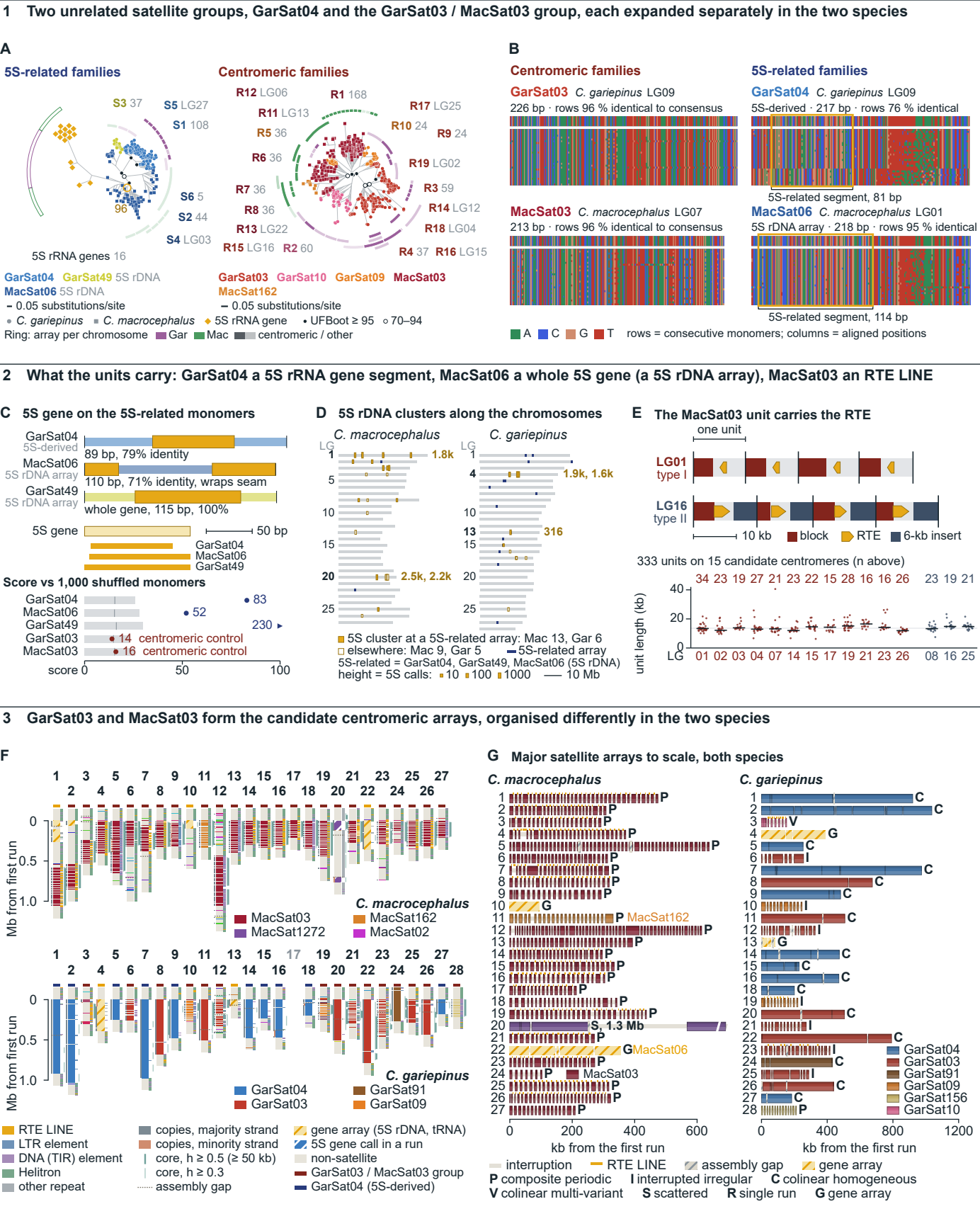

1 Two unrelated satellite groups, GarSat04 and the GarSat03 / MacSat03 group, each expanded separately in the two species
A
5S-related families
Centromeric families
S3 37
S5 LG27
S1 108
S6 5
S2 44
S4 LG03
96
5S rRNA genes 16
R12 LG06
R1 168
R17 LG25
R11 LG13
R10 24
R5 36
R9 24
R6 36
R19 LG02
R7 36
R3 59
R8 36
R14 LG12
R13 LG22
R18 LG04
R15 LG16
R2 60
R4 37
R16 LG15
GarSat04 GarSat49 5S rDNA
MacSat06 5S rDNA
0.05 substitutions/site
GarSat03 GarSat10 GarSat09 MacSat03
MacSat162
0.05 substitutions/site
C. gariepinus
C. macrocephalus
5S rRNA gene
UFBoot ≥ 95
70–94
Ring: array per chromosome
Gar
Mac
centromeric / other
B
Centromeric families
GarSat03 C. gariepinus LG09
226 bp · rows 96 % identical to consensus
MacSat03 C. macrocephalus LG07
213 bp · rows 96 % identical to consensus
5S-related families
GarSat04 C. gariepinus LG09
5S-derived · 217 bp · rows 76 % identical
5S-related segment, 81 bp
MacSat06 C. macrocephalus LG01
5S rDNA array · 218 bp · rows 95 % identical
5S-related segment, 114 bp
A C G T
rows = consecutive monomers; columns = aligned positions
2 What the units carry: GarSat04 a 5S rRNA gene segment, MacSat06 a whole 5S gene (a 5S rDNA array), MacSat03 an RTE LINE
C 5S gene on the 5S-related monomers
GarSat04
5S-derived
89 bp, 79% identity
MacSat06
5S rDNA array
110 bp, 71% identity, wraps seam
GarSat49
5S rDNA array
whole gene, 115 bp, 100%
5S gene
50 bp
GarSat04
MacSat06
GarSat49
Score vs 1,000 shuffled monomers
GarSat04
MacSat06
GarSat49
GarSat03
MacSat03
83
52
230
14 centromeric control
16 centromeric control
score 0
50
100
D 5S rDNA clusters along the chromosomes
C. macrocephalus
C. gariepinus
LG
1.8k
1.9k, 1.6k
316
2.5k, 2.2k
5S cluster at a 5S-related array: Mac 13, Gar 6
elsewhere: Mac 9, Gar 5
5S-related array
5S-related = GarSat04, GarSat49, MacSat06 (5S rDNA)
height = 5S calls: 10 100 1000
10 Mb
E The MacSat03 unit carries the RTE
one unit
LG01
type I
LG16
type II
10 kb
block
RTE
6-kb insert
333 units on 15 candidate centromeres (n above)
34 23 19 27 21 23 22 15 28 16 16 26 23 19 21
unit length (kb)
LG
01 02 03 04 07 14 15 17 19 21 23 26 08 16 25
3 GarSat03 and MacSat03 form the candidate centromeric arrays, organised differently in the two species
F
Mb from first run
C. macrocephalus
MacSat03
MacSat1272
MacSat162
MacSat02
C. gariepinus
GarSat04
GarSat03
GarSat91
GarSat09
RTE LINE
LTR element
DNA (TIR) element
Helitron
other repeat
copies, majority strand
copies, minority strand
core, h ≥ 0.5 (≥ 50 kb)
core, h ≥ 0.3
assembly gap
gene array (5S rDNA, tRNA)
5S gene call in a run
non-satellite
GarSat03 / MacSat03 group
GarSat04 (5S-derived)
G Major satellite arrays to scale, both species
C. macrocephalus
C. gariepinus
S, 1.3 Mb
MacSat162
MacSat06
MacSat03
kb from the first run
GarSat04
GarSat03
GarSat91
GarSat09
GarSat156
GarSat10
interruption
RTE LINE
assembly gap
gene array
P composite periodic
I interrupted irregular
C colinear homogeneous
V colinear multi-variant
S scattered
R single run
G gene array